

VALKENBURG, Netherlands

WMO: 062100

Lat: 52.1397N

Lon: 4.4364E

Elev: 3

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 94-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.2	24.7	14.6	11.5	23.0	19.1	14.4	26.0	34.5	47.7	32.0	48.1	7.9	70	0.613

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.9	80.9	67.6	76.8	65.8	73.3	64.2	69.2	77.7	67.3	74.3	65.7	71.6	9.1	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.3	96.9	72.5	64.6	91.3	70.6	63.2	86.6	68.8	33.2	78.1	31.7	74.3	30.5	72.0	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.2	25.6	22.7	DB	16.5	88.3	5.9	2.6	12.3	90.2	8.8	91.8	5.5	93.3	1.2	95.2	(4)
(5)			WB	15.8	73.1	5.8	1.6	11.6	74.2	8.2	75.1	5.0	76.0	0.7	77.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.9	39.0	39.8	43.1	48.7	54.7	59.4	63.7	63.8	59.0	52.8	46.1	40.1		
	DBStd	10.64	7.33	6.64	5.54	6.09	5.68	4.97	4.83	4.37	4.21	5.41	5.88	7.45		
	HDD50	1449	341	289	221	96	17	0	0	0	0	34	140	311		
	HDD65	5247	805	707	678	490	324	181	84	76	185	380	566	771		
	CDD50	1783	1	2	7	57	162	282	425	428	271	119	24	5		
	CDD65	105	0	0	0	0	4	13	44	39	5	0	0	0		
	CDH74	660	0	0	0	4	43	103	267	218	24	1	0	0		
	CDH80	137	0	0	0	0	5	21	63	46	2	0	0	0		
	WSAvg	10.5	12.4	12.1	11.5	10.0	10.2	10.0	9.4	8.9	9.3	10.1	10.5	11.8		
	PrecAvg	36.5	3.1	2.7	2.1	1.7	2.2	2.7	3.3	4.1	3.6	3.6	4.0	3.5		
	PrecMax	46.9	6.7	5.2	4.7	4.1	4.5	5.9	7.7	9.2	9.8	6.3	7.5	6.3		
(15) Precipitation	PrecMin	26.3	0.4	0.8	0.4	0.0	0.8	0.9	0.8	0.7	0.5	0.7	0.3	0.7		
	PrecStd	5.0	1.5	1.2	1.1	1.0	1.1	1.5	1.8	2.4	2.1	1.6	1.7	1.4		
	DB	54.4	55.9	62.7	73.3	79.0	83.2	86.1	84.9	77.1	69.2	61.0	56.3			
	MCWB	51.3	49.7	53.7	58.4	63.6	68.3	69.6	69.3	66.9	62.6	57.5	53.5			
	2% DB	51.9	52.2	57.3	67.0	73.8	76.6	81.0	80.1	72.4	64.7	58.0	54.0			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	49.6	48.6	51.1	56.5	61.0	65.1	67.7	68.0	64.7	60.5	55.3	51.7			
	DB	49.8	50.3	53.8	62.5	69.0	71.9	76.4	75.3	68.8	62.5	56.2	52.0			
	MCWB	47.6	47.4	49.2	54.0	59.2	63.2	66.6	66.2	62.7	59.0	53.7	49.9			
	10% DB	48.0	48.4	51.2	58.2	64.5	68.3	72.4	71.5	66.1	60.6	54.1	49.9			
	MCWB	46.1	46.0	47.5	52.1	57.1	61.1	64.7	64.6	61.2	57.4	51.6	47.6			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	51.9	51.4	55.1	60.1	66.6	70.1	72.3	71.7	67.9	63.4	57.8	54.1			
	MCDB	53.7	54.1	61.0	70.1	76.2	80.0	80.6	81.7	74.7	67.7	59.5	55.7			
	2% WB	49.9	49.3	51.9	57.2	62.4	66.4	69.2	69.0	65.5	61.1	55.8	52.0			
	MCDB	51.8	51.5	55.9	65.2	69.9	73.7	77.9	76.8	71.0	64.0	57.8	53.6			
	5% WB	47.9	47.7	49.7	54.9	60.0	64.0	67.3	67.0	63.7	59.6	53.9	50.2			
(33) Mean Daily Temperature Range	MCDB	49.4	49.8	53.1	61.3	67.3	70.3	74.4	73.4	67.8	62.1	55.8	51.9			
	10% WB	46.3	46.2	47.9	52.7	57.8	61.9	65.5	65.5	62.0	57.9	51.9	48.0			
	MCDB	47.8	48.2	50.5	57.3	63.8	67.1	71.4	70.7	65.4	60.2	53.8	49.8			
	5% DB	7.7	8.7	10.9	13.5	13.4	13.1	12.5	12.9	12.2	10.8	8.6	7.7			
	MCDBR	7.7	9.7	15.8	20.2	21.5	20.4	19.7	19.5	16.4	12.2	9.0	7.3			
(36) Clear-Sky Solar Irradiance	MCWBR	7.2	7.5	10.7	11.5	11.6	11.2	9.8	9.6	9.9	8.9	7.8	7.1			
	5% WB	7.4	8.5	13.7	18.2	19.0	17.7	17.3	16.7	14.6	11.7	8.6	7.6			
	MCDBR	7.3	7.2	9.9	10.9	11.2	10.7	9.4	9.2	9.5	9.3	7.8	7.5			
	taub	0.342	0.362	0.395	0.418	0.424	0.419	0.428	0.419	0.397	0.377	0.354	0.337			
	taud	2.312	2.305	2.231	2.213	2.257	2.287	2.273	2.312	2.361	2.367	2.333	2.306			
(42) All-Sky Solar Radiation	Ebn at Noon	203	234	249	259	263	265	260	256	247	226	195	181			
	Edh at Noon	21	27	35	40	41	40	40	36	31	25	20	18			
	RadAvg	222	440	827	1338	1625	1735	1644	1380	998	571	258	167			
(45)	RadStd	23	81	99	131	110	118	175	112	85	49	24	19			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (80 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page